

Flexibility training does not increase behavioral diversity of Florida scrub-jays

McCune, KB^{1,2}

Barve, S³

2026-07-26



Affiliations:

1) Institute for Social, Behavioral and Economic Research, University of California Santa Barbara, Santa Barbara, CA, USA

2) College of Forestry, Wildlife and Environment, Auburn University, Auburn, AL, USA

3) Archbold Biological Station, Venus, FL, USA.

Corresponding author: kelseybmccune@gmail.com

This is a post-study manuscript of the programmatic [registered report](#) that has been pre-study peer reviewed and received an In Principle Acceptance on 8 Sep 2022 by:

Chris Chambers (2022) The role of behavioural flexibility in promoting resilience to human environmental impacts. *Peer Community in Registered Reports*. <https://rr.peercommunityin.org/articles/rec?id=200>.

Reviewers: Gloriana Chaverri, Vedrana Šlipogor, and Alizée Vernouillet



See the reproducible manuscript ([Rmd](#)) version for the code

ABSTRACT

Human modifications of environments are increasing, causing global changes that other species must adjust to or suffer from. Behavioral flexibility (hereafter ‘flexibility’) could be key to coping with rapid change. Behavioral research can contribute to conservation by determining which behaviors can predict the ability to adjust to human modified environments and whether these can be manipulated. Most research that seeks to change animal behavior to facilitate conservation primarily focuses on one specific behavior to address one potential threat (e.g., response to a novel predator). However, training a domain general cognitive ability, such as flexibility, has the potential to change a whole suite of behaviors, which could have a larger impact on adjusting to human modified environments. This study asked whether flexibility can be increased by experimentally increasing environmental heterogeneity and whether such an increase helped species succeed in human modified environments. We aimed to 1) conduct flexibility interventions in disturbance-resilient species that are successful in human modified environments (California scrub-jays or blue jays) to understand how flexibility relates to success; and 2) implement these interventions in a disturbance-sensitive species (Florida scrub-jays) to determine whether flexibility as a generalizable cognitive ability can be trained and whether such training improves success in human-modified environments. Due to funding and time constraints, we were only able to conduct this research in Florida scrub-jays from two sites: fire-maintained, unfragmented habitat at Archbold Biological Station (“natural”) and suburban neighborhoods. We found no difference in behavioral flexibility before the intervention between jays in the natural or suburban sites. We were then able to successfully increase behavioral flexibility of wild jays through serial reversal learning, but we found no effect of this intervention on foraging or habitat use behaviors. Because we were not able to test successful species, we instead conducted an unregistered analysis comparing foraging and habitat use behavior between suburban and natural site Florida scrub-jays. Suburban jays used a greater variety of habitats than Archbold jays but there was no difference in foraging behavior. This research demonstrates that behavioral flexibility may not relate to adaptation to environmental change in this species, but some jays are able to persist in human modified environments and use the greater diversity of microhabitats present there. To better understand the role of behavior in the ability of Florida scrub-jays to adapt to environmental change, future work should evaluate the variation in other relevant traits like boldness, exploration and movement behavior.

REGISTERED REPORT DETAILS

- **Level of bias = 6:** This registered report was written (Jul 2021-May 2022), and revised after round one of peer review at Peer Community in Registered Reports (Jul 2022) prior to collecting any data.
- **Programmatic registered report:** This is one of three Stage 2 articles that will result from the original, comprehensive Stage 1 registered report: one for toutouwai, one for grackles, and one for jays (this current manuscript).
- **Deviations from the Stage 1 registered report:**
 - We were unable to sample disturbance-resilient blue jays or California scrub-jays and so we cannot address J.Q2 (one of four specific research questions) at this time.
 - To begin to test the question of whether behavior relates to success in human-modified environments, we conducted an *unregistered analysis* comparing behavior of Florida scrub-jays (a species that is declining due to anthropogenic change and therefore disturbance-sensitive) that were persisting in suburban areas to those in natural habitats managed with fire to promote jay population growth.

INTRODUCTION

Human modified environments are expanding (Goldewijk, 2001; X. Liu et al., 2020; Wu et al., 2011), causing global changes that other species must adjust to or suffer from (Alberti, 2015; Chejanovski et al., 2017;

Ciani, 1986; Federspiel et al., 2017). Behavioral flexibility (hereafter ‘flexibility’) could be key for adjusting to such change: individuals interact with their environment through behaviors such as foraging and habitat selection, making a focus on the adaptability of these behavioral decisions crucial to understanding how species adjust to environmental changes (Lee & Thornton, 2021). One of the top priorities for behavioral research to maximize conservation progress is to determine which cognitive abilities and behaviors can predict the ability to adjust to human modified environments and whether these can be manipulated (Moseby et al., 2016). The rare research that manipulates behavior in a conservation context usually focuses on training specific behaviors (for example, predator recognition through predator exposure) to improve individual success in the wild (Jolly et al., 2018; Moseby et al., 2012; Ross et al., 2019; West et al., 2018; see review in Tetzlaff et al., 2019). However, training a general cognitive ability, such as flexibility – the ability to rapidly adapt behavior to changes through learning throughout the lifetime (see the theory behind this definition in Mikhalevich et al., 2017) – has the potential to change a whole suite of behaviors and more broadly influence success in adjusting to human modified environments. Recent evidence supports this hypothesis: as far as we are aware, our previous research in the great-tailed grackle was the first to show that flexibility can be manipulated using serial reversal learning of color preferences, and that the manipulated individuals showed more diverse foraging behavior (Logan et al., 2025), were faster to change behavior in a different experimental context (locus switching on a puzzlebox) as well as being more innovative (solved more loci on a puzzlebox) (Logan et al., 2023).

Environments where informational cues about resources vary in a heterogeneous (but non-random) way across space and time are hypothesized to open a pathway for species to functionally detect and react to such cues via flexibility (Mikhalevich et al., 2017). Human modified environments likely provide a different set of informational cues that vary heterogeneously across space and time, and the species that are successful in such environments are likely those who are able to detect and track such cues. Globally, bird species with more diverse foraging behavior in the wild were less likely to be threatened with extinction (Ducatez et al., 2020), supporting the idea that the ability to flexibly adapt to new foraging challenges reduces population decline. Because heterogeneous environments are hypothesized to select for flexibility (Wright et al., 2010), we expect that exposing individuals to experimentally increased environmental heterogeneity will train individuals to be more flexible, which will then increase their success in such environments (Fig. 1). Success can relate to any number of variables regarding the usage of and investment in resources and response to threats, from improved foraging efficiency to increased dispersal and survival within human modified environments, to placing nests in more protective locations. Whether a measure of success is predicted to relate to flexibility depends on what is already known about the particular population and their particular environment.



Figure 1. The theory behind this research illustrated by a directed acyclic graph (DAG), which is a theoretical model of the causal relationships among the key variables in our investigation. Based on the theoretical background provided by Mikhalevich et al. (2017), we assume that more heterogeneity causes more flexibility, which then causes more success in human modified environments.

This investigation asked whether a method of experimentally increasing environmental heterogeneity through serial reversal learning leads to increased behavioral flexibility and greater fitness in human modified environments. Serial reversal learning tasks have been performed with a wide diversity of species (birds: Bond et al., 2007; bumblebees: Strang & Sherry, 2014; stingrays: Daniel & Schluessel, 2020). There is variation across individuals and species in their performance, however almost all previous studies show that individuals improve the time it took to change their behavior if the reversal manipulation is given multiple times in sequence (rats: Mackintosh et al., 1968; guppies: Lucon-Xiccato & Bisazza, 2014; poison frogs: Y. Liu et al., 2016; grackles: Logan et al., 2023). We aim to conduct a flexibility manipulation in common species

that are resilient to human-disturbance (California scrub-jays or blue jays) to understand how flexibility relates to success, and implement these interventions in a disturbance-sensitive species (Florida scrub-jays). Ultimately, we were unable to conduct the flexibility manipulation in a disturbance-resilient jay species. Nevertheless, our previous research in disturbance-resilient great-tailed grackles already demonstrated that increased flexibility increases the diversity of behaviors related to success in human modified environments (Logan2025behavioral?). What is still unknown, is whether this manipulation can also influence the success of a species that is threatened or endangered with extinction. That is the research we focused on here.

While we did not examine the potential spread of the post-manipulation success behaviors from manipulated individuals to individuals that are not involved in our studies, we acknowledge that this is a possibility worthy of future investigation. Manipulating the flexibility of a few individuals could have population-level effects because significant research on social information use in birds (e.g., Valente et al., 2021) demonstrates the potential for the manipulated behavior to disseminate to conspecifics (for example, if manipulated individuals are faster at locating new resources, which could attract the attention of conspecifics, or if unmanipulated individuals copy the manipulated individuals' nesting or foraging locations). Indeed, there is evidence for social learning in California and Florida scrub-jays (K. B. McCune et al., 2022; Midford et al., 2000). Even if the behaviors of manipulated individuals do not spread in the population through social learning, investing in the training of specific individuals to increase their success in the wild could still have conservation impacts. In some cases, it is possible to train many individuals in a population or a species because there are not many individuals left (Greggor et al., 2021). It is also possible to train all individuals involved in a conservation management event such as a translocation (Greggor et al., 2021). Therefore, there can still be significant population consequences even if each individual needs to be trained to achieve the goal.

While our previous research on great-tailed grackles illustrated that flexibility can be increased and can affect behavioral interactions with the environment, the research presented here took the next step by evaluating the generalizability of this technique to a distinct species negatively affected by anthropogenic change. Consequently, the results from this research are important to advance our understanding of the causes and consequences of flexibility by linking behavior to environmental change, cognition, and conservation of declining species.

RESEARCH QUESTIONS

Can behavioral flexibility in individuals be increased by increasing environmental heterogeneity? If so, does increased flexibility help individuals succeed in human modified environments?

Prediction 1: Flexibility can be increased in individuals and such an increase **improves the likelihood of success in human modified environments**. This would indicate that the abilities involved in tracking changing resources in the environment are the same as or related to the abilities involved in succeeding in human modified environments. It would also indicate that flexibility is trainable and that such training could be a useful conservation tool for threatened and endangered species.

Prediction 1 alternative 1: Flexibility can be increased in individuals, but such an increase **does not improve the likelihood of success** in human modified environments. This would indicate that species associated with human modified environments form this association for reasons other than their flexibility, and that threatened species are likely not very successful in human modified environments for reasons unrelated to their ability to change their behavior with changing circumstances. An alternative could be that the changes induced by the increase in flexibility do not persist for sufficiently long times to make a difference on the subsequent likelihood of success (changes in great-tailed grackles were still present for four weeks after the manipulation and longer time periods were not attempted so the threshold is unknown Logan et al., 2022).

Prediction 1 alternative 2: Flexibility can be increased in some populations, but not others. This would

161 indicate that **flexibility manipulations may not work for all populations**, and that the effectiveness
162 of such experiments should first be tested in the population of interest before including such an intervention
163 in a conservation plan. If flexibility is not manipulatable in threatened populations, this would indicate that
164 they are likely not very successful in human modified environments because of their inability to change their
165 behavior with changing circumstances, and that flexibility is not trainable. If flexibility is not manipulatable
166 in populations that are successful in human modified environments, this could indicate that they might have
167 used flexibility in the past when originally forming the association, but the need to maintain flexibility in
168 their repertoire is no longer necessary (Wright et al., 2010). In populations where flexibility is not manipu-
169 latable, this would indicate that the abilities involved in tracking changing resources in the environment are
170 independent of the abilities involved in succeeding in human modified environments.

Population-specific background and tailored research questions

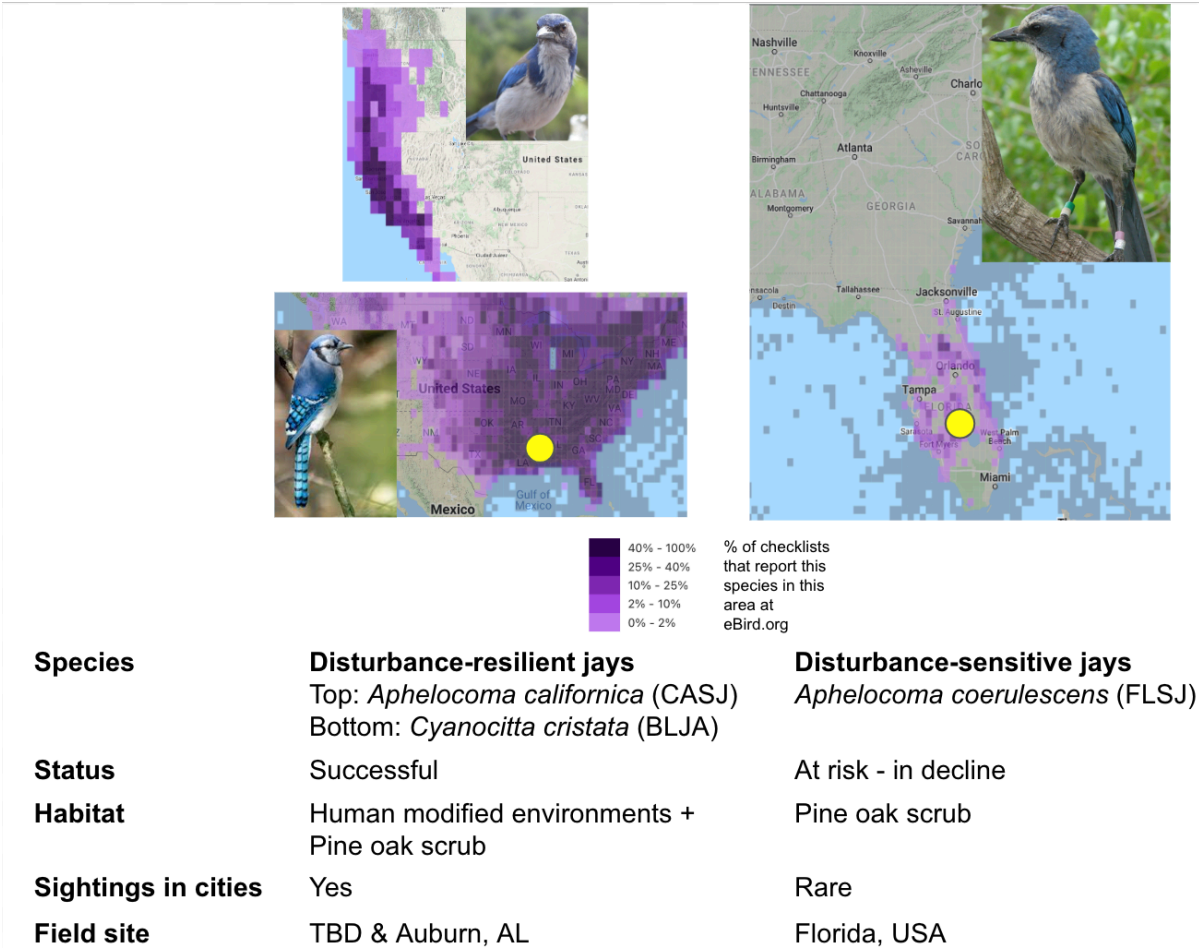


Figure 2. Comparing the species involved in this investigation relative to their geographic range and association with human modified habitats. The yellow dots represent field site locations. Species range data comes from eBird (www.ebird.org). Photo credit: CASJ, Corina Logan; BLJA, Rhododendrites; FLSJ, VvAndromedavV.

Background

Jay species exhibit a diversity of social systems and success in colonizing suburban and urban areas (Fig. 2). California scrub-jays (*Aphelocoma californica*, hereafter “CASJ”) and blue jays (*Cyanocitta cristata*, hereafter “BLJA”) are singular, monogamous breeders that are increasing in abundance, expanding their range sizes, and highly successful in natural, suburban, and urban areas (Blair, 1996; Curry et al., 2017). We therefore consider these “disturbance-resilient” (DR) jay species. In contrast, the Florida scrub-jay (*A. coerulescens*; hereafter “FLSJ”) is a “disturbance-sensitive” (DS) jay species that is threatened, endemic, and range-restricted to xeric oak scrub habitat in Florida (Woolfenden & Fitzpatrick, 1996). Some FLSJ can persist in suburban (but not urban) habitats after conversion from xeric oak scrub, however suburban populations of FLSJ steadily decline (Bowman pers. comm.). This differential adaptation to human-dominated areas across species could be related to behavioral flexibility in habitat use and foraging breadth.

All three species forage primarily on mast (acorns, hazelnuts, etc.) that they cache throughout their territory, which makes it available to eat year-round. They are also opportunistic omnivores and specifically need high-fat and high-protein arthropods to feed to nestlings and fledglings (Curry et al., 2017). Nesting and foraging substrates can be drastically different in human modified environments compared to natural areas [e.g., predominance of non-native vegetation; Tuomainen & Candolin (2011)]. For example, fire suppression in suburban areas of Florida reduces habitat quality for FLSJ (Woolfenden & Fitzpatrick, 1996), potentially leading to higher rates of brood reduction through predation or nestling starvation related to a lack of nutritionally complete prey items (Shawkey et al., 2004). The ability to flexibly adjust behavior in human modified environments to compensate for these habitat and foraging changes could thus influence survival and reproductive success.

We compared behavioral flexibility of Florida scrub-jays, between suburban and natural populations to determine whether variation in flexibility relates to variation in presence in these habitats. Subsequently, we tested whether a serial reversal learning manipulation to increase flexibility increased the foraging and microhabitat breadth of jays in suburban and natural environments. Although we do not quantify social diffusion of the manipulated behaviors here, previous research demonstrates that this species has the capacity to socially learn foraging information discovered by others and so manipulating the flexibility of a subset of individuals has the potential to affect the population (Midford et al., 2000).

Tailored Research Questions

For all research questions, Table 1 summarizes our predictions, analysis plans, interpretations for the various directions the results could go, and the hypotheses that could be contradicted given the various outcomes. Table 2 summarizes the achieved species, sample sizes and results.

- **J.Q1: Do jay populations in human modified areas differ in baseline behavioral flexibility compared to populations in natural areas?** We investigated this question by comparing performance on serial reversal learning in the wild between Florida scrub-jays in natural areas and jays in human modified areas.
- **J.Q2: Are disturbance-resilient (DR) jays more behaviorally flexible than disturbance-sensitive (DS) jays?** We aimed to investigate this question by comparing performance on reversal learning in the wild between DR and DS jay species. However, we were unable to test J.Q2 because it was not possible to sample CASJ or BLJA within the timeframe of this study.
- **J.Q3: Does manipulating behavioral flexibility alter the number of microhabitats used?** We investigated this question by tracking their presence in a variety of microhabitats before and after manipulating their flexibility using serial reversal learning in the wild (Fig. 3). We only count that a microhabitat was used if the individual had at least 5% of their data points there. This prevents a microhabitat from being counted even if an individual was simply moving through it, and therefore not necessarily using it.
- **J.Q4: Does manipulating flexibility alter the number of different food items taken by jays?** We investigated this question by tracking the various food items they take before and after manipulating their flexibility using serial reversal learning in the wild (Fig. 3).

Do flexibility manipulated individuals differ in how they interact with their environment?

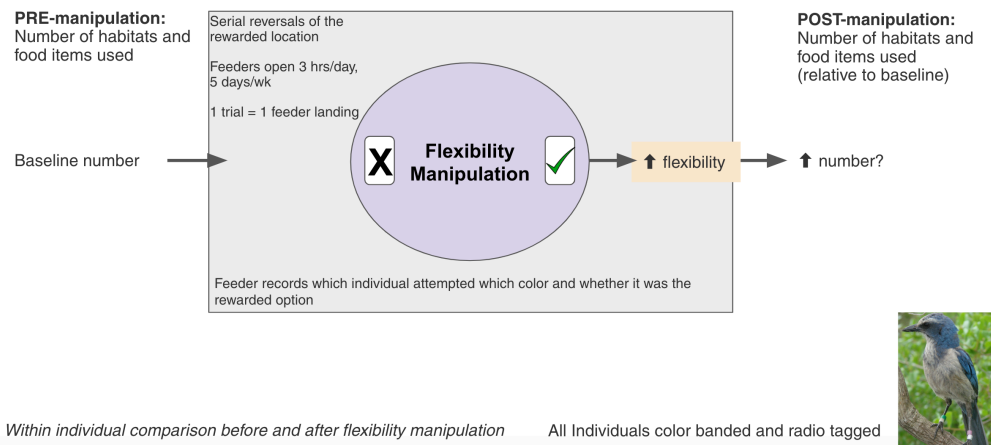


Figure 3. The reversal learning experiment in a group context (Design 2 from the ManyIndividuals registered report) tailored to the jay research questions. The white rectangles represent feeder locations, the feeder with the X is in the unrewarded location while the feeder with the green check is the rewarded location.

Table 1. Study design for the jay research. Note that we are unable to test Q2 because it was not possible to sample CASJ or BLJA within the timeframe of this study. References that were not already cited in the introduction: Galbraith et al. (2015), Lapiedra et al. (2017), Rice et al. (2003), Emery & Clayton (2004), Sol et al. (2002), Peterson et al. (2011), Grinnell (1917).

Question	Hypothesis	Sampling plan	Analysis plan	Rationale for deciding test sensitivity to confirm/disconfirm hypothesis	Interpretation given different outcomes	Theory that could be shown wrong by the outcomes
1. Do jay populations in human modified areas differ in baseline behavioral flexibility compared to populations in natural areas?	Prediction 1.1: Suburban jays are more flexible than jays in natural areas	Simulations using bespoke Bayesian models in Logan et al. (2021) showed a high likelihood of detecting	Bayesian model: Response: phi and lambda Explanatory: Habitat (suburban/ natural) (see Analysis Plan)	Contrasts will determine whether the before and after conditions differed from each other. We will conclude there is a difference if the confidence interval does not cross zero.	This implies that flexibility is related to the ability to occupy human modified environments where spatial and temporal heterogeneity of resources is high.	Selection for exploitation of supplementary food (Galbraith et al. 2015), where individuals learn to depend on anthropogenic food sources and are less likely to flexibly sample alternative resources.
	Prediction 1.2: Suburban jays are less flexible than jays in natural areas	differences with a sample size of 15 when mean differences in phi were at least 0.01 and lambda at least 3	(see Analysis Plan)	False positives: the power analyses suggest that false positives are unlikely even with small sample sizes.	This implies that human modification of the environment has led to less spatial and temporal heterogeneity of resources. For example, the prevalence of bird feeders in suburban areas leads to consistently available food.	(1) Flexibility facilitates adapting to environmental change (see Introduction)
	Prediction 1.3: There is no difference in flexibility between suburban jays and jays in natural areas	(see Analysis Plan)	(see Analysis Plan)	False positives: the power analyses suggest that false positives are unlikely even with small sample sizes. Accordingly, we will interpret any contrast that does not cross zero as indicating an effect.	This implies that additional behavioral (e.g. boldness, Lapiedra et al., 2017) or genetic traits may facilitate success in human modified environments.	(1) The urban filter (Lapiedra et al. 2017) where novel anthropogenic pressures select for flexible individuals.
2. Are disturbance resilient (DR) jays more behaviorally flexible than disturbance sensitive (DS) jays?	Prediction 2.1: DR jays are more flexible than DS jays		Bayesian model: Response: phi and lambda Explanatory: Species (see Analysis Plan)	False negatives: the power analyses suggest that, especially with small sample sizes, we will not have sufficient power to exclude the possibility that an effect exists even though our model does not indicate an effect (contrast crosses zero).	This difference may explain the range expansion and greater success of DR jays in human modified environments.	Range expansion instead relates to ecological niche differentiation (Rice et al. 2003), where DR jays evolved to occupy a niche that more closely resembles human modified environments than DS jays.
	Prediction 2.2: DR jays are less flexible than DS jays		(see Analysis Plan)	False negatives: the power analyses suggest that, especially with small sample sizes, we will not have sufficient power to exclude the possibility that an effect exists even though our model does not indicate an effect (contrast crosses zero).	This implies that flexibility is not related to success in human modified environments and that flexibility may instead be related to a different, unknown social or environmental characteristic. For example, the cooperative breeding system of the DS jay species, the Florida scrub-jay, may favor increased flexibility for responding to group mates' behavior.	(1)
	Prediction 2.3: DR jays and DS jays are equally flexible		(see Analysis Plan)	False negatives: the power analyses suggest that, especially with small sample sizes, we will not have sufficient power to exclude the possibility that an effect exists even though our model does not indicate an effect (contrast crosses zero).	This implies flexibility is not related to success in human modified environments and the level of flexibility is potentially an evolutionary conserved trait from a corvid common ancestor (Emery & Clayton 2004).	(1)
3. Habitat: Does manipulating behavioral flexibility alter the number of microhabitats used?	Prediction 3.1: Flexibility can be increased and such an increase alters daily habitat use to include more variety of habitats	Simulations using bespoke Bayesian models showed a high likelihood of detecting differences with a sample size of 20 when mean difference in the proportion of microhabitats used was at least 0.1 (standard deviation=0.1) or 0.15 (SD=0.2)	Bayesian model: Response: Number of microhabitats used per individual Explanatory: Condition (before/after) Random: Condition ID (see Analysis Plan)		Increasing behavioral flexibility, with a serial reversal learning manipulation, increases the likelihood the individual will sample new areas while foraging.	(2) Habitat preferences and the foraging niche are fixed within species because each species evolves within a specific ecological niche (Grinnell 1917; Peterson et al. 2011).
	Prediction 3.2: Flexibility can be increased and such an increase alters daily habitat use to decrease the variety of habitats used		(see Analysis Plan)		Increasing behavioral flexibility potentially leads to increased foraging breadth or use of resources within one habitat, rather than leading to sampling across habitat types.	Behavioral flexibility facilitates the use of novel habitats and invasion success through dietary generalism (Sol et al. 2002)
	Prediction 3.3: Flexibility can be increased but has no effect on the variety of habitats used		(see Analysis Plan)		This suggests that the cognitive ability behavioral flexibility may not generalize to all domains (e.g. may relate to foraging but not habitat use). Alternatively, it could indicate that they might have used flexibility in the past when originally forming the association, but the need to maintain flexibility in their repertoire is no longer necessary, or that changes induced by the increase in flexibility do not persist for sufficiently long times to make a difference on the subsequent likelihood of success.	(1) (3) Behavioral flexibility is a general cognitive ability (see Introduction)
4. Foraging: Does manipulating flexibility alter the number of different food items taken by jays?	Prediction 4.1: Flexibility can be increased and such an increase alters daily foraging breadth to include more variety of food items	Simulations using bespoke Bayesian models showed a high likelihood of detecting differences with a sample size of 20 when mean differences in the number of foods taken were at least 1 (and a standard deviation of 2)	Bayesian model: Response: Number of foods taken per individual Explanatory: Condition (before/after) Random: Condition ID (see Analysis Plan)		Increasing behavioral flexibility, with a serial reversal learning manipulation, increases the likelihood the individual will sample new food sources while foraging.	(2)
	Prediction 4.2: Flexibility can be increased and such an increase alters daily foraging breadth to decrease the variety of food items taken		(see Analysis Plan)		Increasing behavioral flexibility potentially leads to increased use of the same type of food items across habitat types, rather than leading to sampling more food items within one habitat.	Behavioral flexibility facilitates the use of novel foods through sampling new foods and foraging strategies (Sol et al. 2002)
	Prediction 4.3: Flexibility can be increased but has no effect on foraging breadth		(see Analysis Plan)		This suggests that the cognitive ability behavioral flexibility may not generalize to all domains (e.g. may relate to habitat use but not foraging breadth). Alternatively, it could indicate that they might have used flexibility in the past when originally forming the association, but the need to maintain flexibility in their repertoire is no longer necessary, or that changes induced by the increase in flexibility do not persist for sufficiently long times to make a difference on the subsequent likelihood of success.	(1) (3)

METHODS

Planned sample

All data were collected from September - November 2022 and October 2023 - February 2024. We studied Florida scrub-jays (FLSJ) in unfragmented, fire-maintained habitat (~1,000 acres) at Archbold Biological Station (natural areas), as well as in the adjacent suburbs of Lake Placid (human modified areas) in south central Florida, USA. Much of the study population at Archbold Biological Station was already individually

identifiable with color leg bands. We caught an additional 24 jays using walk-in traps and mist nets; collected their biometric data, and a blood sample; applied colored leg bands; and released them at their point of capture. While blood and biometric data were not directly used in this study, these methods contributed to the long-term genetic and demographic information maintained by the Avian Ecology Lab at Archbold.

We collected data on pre- and post-manipulation success measures (foraging and habitat use, see below) and conducted the experiment within the non-breeding season to control for potential temporal differences in the environment and behavior. We planned to use radio tags to find jays for observing before and after manipulation success measures. However, FLSJ are highly territorial, and we were always able to find the jays to collect observational data without the radio tags.

To determine whether the flexibility manipulation has influenced the ability of jays to persist in human modified environments, we planned for half of the total sample of jays from areas with access to human-supplemented food (i.e. private property, university campus, parks adjacent to neighborhoods with feeders) and the other half from natural areas. To address Q1, testing for differences in baseline behavioral flexibility between jays in natural or human-modified environments, the power analysis (see Supplementary materials S1) showed a high likelihood of detecting differences with a minimum sample of 15. We were able to quantify baseline behavioral flexibility in 14 FLSJ (8 in natural areas, 6 in human-modified areas; Table 2).

To address Q3 and Q4, we tracked baseline behavior and changes after the flexibility manipulation via all-occurrence sampling during focal observations (Altmann, 1974) that last for 60 min. During observations, we documented all occurrences of microhabitats used within the territory and foraging behavior (see Observational methods, below). Our planned minimum sample size was 20 individuals with a minimum of 4, 60-minute observations per individual (at least 2 per month) pre-manipulation and a minimum of 4 per individual (at least 2 per month) post-manipulation (at least 320 minutes of observations in total). Behavioral observations occurred at least 24 hours apart. In total, we collected behavioral observations on 30 FLSJ (19 in natural areas, 11 in human modified areas). We met or exceeded the minimum observation sessions (mean = 7.6, range = 3-14 observations) before and after the manipulation. There was one exception where an individual disappeared after the manipulation and was not found during extensive searching of the territory and surrounding areas and so was presumed dead. We only collected one-third of the necessary post-manipulation data on this individual. However, we were only able to complete the serial reversal learning manipulation on 9 jays (8 natural, 1 suburban; Table 2).

Observational methods

During observations, we recorded each microhabitat the individual is present in and all food items consumed. Before data analysis, to ensure that we were only including the microhabitats individuals use (rather than just pass through), we filtered the data to include only microhabitats that account for at least 5% of their data points. Although we may not observe every possible microhabitat or food item the individual may use, by equally sampling before and after the manipulation we can detect changes in habitat use. We counterbalanced observations across the morning (0600-1159) and afternoon (1200-1600) for all individuals in the study. In this way, we collected a random sample of data from active and inactive time periods for all individuals.

Microhabitat types in the suburban habitat (<100m from human structure) included: vertical human structure (e.g. building, bench), native vegetation, non-native vegetation, grass, impervious surface, and dirt. In the natural habitat (>100m from human structure), microhabitat types included all previous categories, but not human structure or impervious surface. All categories were further defined by whether the subject was high (>3m) or low (<3m). For example, grass and impervious surfaces can occur above 3m if grass is on the roof of a building, and if an individual was walking on the impervious surface of an upper floor parking garage.

Food types were broken down into plant (seed, fruit, human-provided, or unknown plant) and animal (insect larva, adult insect, amphibian, crustacean, reptile, mammal, bird, egg, human-provided, or unknown animal). “Human-provided” indicates any food item that was acquired from a store at some point and is left out by humans. For example, sunflower seeds were considered human-provided if they were in the form of bird seed

or a human snack. Sunflower seeds were only counted in the “seed” category if the bird is seen eating it from a plant.

Additionally, for all observations we used binoculars so that we remained far enough from the focal individual that our presence was not affecting their behavior. That distance can be different for each individual and species. However, if at any point the focal individual showed that it was affected by our presence, by looking directly at the researcher, alarm calling or startling, then we ended the observation immediately, dropped all data from that observation session and attempted the observation again the next day.

Behavioral Flexibility

Prior to measuring behavioral flexibility, we collected the 4 minimum observation sessions to determine baseline values for microhabitat use and foraging breadth. Our approach for all behavioral flexibility methods involved individual jays participating in a spatial reversal learning task in the wild. We set up identical remote-controlled feeders at multiple study sites, each containing several jay territories, spaced at least 2km apart: 2 sites in natural habitat, 2 in human-modified habitat. Feeders were only present on the territory for jays during habituation and testing.

Feeder habituation

One feeder was set up at each location that contained food and the experimenter used the remote to feed the focal individual every time it approached. Habituation consisted of 2-hour sessions between 0800-1500 on a minimum of 4 consecutive days or until at least 1 banded jay per territory had visited the feeder at least 3 times without showing fear behaviors such as hesitation to approach or touch the feeder, and jumpiness.

Baseline behavioral flexibility

We measured baseline behavioral flexibility as the pre-manipulation performance on reversal learning of the spatial location of the rewarding feeder. We set up two identical remote-controlled feeders 5m apart which both contained one type of high value food (i.e., peanuts). Spatial location was the only cue indicating which feeder would dispense food because both feeders were opaque and contained peanuts, eliminating visual or olfactory differences. If a feeder needed to be refilled, we refilled all feeders consecutively in the same time period and for the same amount of time even if that feeder did not need much or any food (in these cases, we pretended to fill the feeder). This eliminated potential confounds in jay performance from cues provided by a differential amount of experimenter attention given to the feeders during refilling. Reversal learning sessions for baseline behavioral flexibility (and the manipulation experiment, see below) consisted of 30-120 min sessions per day per territory, up to 4 days per week.

The reversal learning experiment involves an initial stage of associative learning (referred to as “Reversal 0”) where one of the two feeders, at a consistent location (north or south, east or west) within the territory, was randomly selected to dispense food when the focal jay approached. If the subject visited the non-rewarded feeder, this incorrect choice was recorded, but the feeder did not open. Each visit by the focal jay to the correct or incorrect feeder was considered a trial. Jays passed this stage when they correctly chose the rewarded feeder in at least 10 trials out of the most recent 12 trials (see S1 of the supplementary materials for details on the simulations that determined that threshold). At this point, jays had an expectation of where they could receive food, so we switched which feeder dispensed food to test their behavioral flexibility as the speed that they reversed the previously learned association. This first reversal stage (“Reversal 1”) is the measure of baseline behavioral flexibility before it is increased through the serial reversal learning manipulation. Jays similarly passed Reversal 1 when they chose correctly on 10 out of the most recent 12 trials.

Flexibility manipulation (Design 2 from the registered report)

After jays passed Reversal 1, we switched to rewarding jays at the other feeder location to initiate the serial reversal learning flexibility manipulation stage. For each subsequent reversal (Reversal 2, 3, etc.) the criteria to pass was again correctly choosing the rewarded feeder in at least 10 out of the most recent 12 trials. Serial reversals continued until jays passed two consecutive reversals in 12 trials or fewer (see S2 of the supplementary materials for details on the simulations that determined that threshold). Given that the reversal passing criteria is 10 correct choices out of the most recent 12 trials, jays must pass two reversals as fast as possible to be considered to have increased their behavioral flexibility.

After the manipulation was complete for the focal individual, we again conducted the all-occurrences focal observation sessions to measure microhabitat use and foraging breadth. We began collecting post-manipulation data on an individual as soon as it passed the manipulation because it is unknown for how long any potential effects of the manipulation will last.

For more social species, like the FLSJ, who may experience the experiment in groups, there is the potential that individuals who were not in the manipulation condition or who have not yet passed the manipulation condition will learn about post-manipulation success behaviors from the manipulated individuals who passed the experiment. We are ultimately interested in determining whether we can change success behaviors of threatened populations or species as a result of the flexibility manipulation. If part of this change is the result of social learning from some of the manipulated individuals, it will still result in a change even if we do not quantify what percentage of the mechanism comes from individual learning as a result of the manipulation or social learning after the manipulation. If there is a change in success measures between before and after the manipulation, the manipulation will have been the cause in either case.

Open data

The data are published in the Knowledge Network for Biocomplexity's data repository ([doi:10.5063/F19885J8](https://doi.org/10.5063/F19885J8)).

Protocols

- Jay [protocol](#) and [data sheet templates](#)
- [Protocol for applying radio tags and conducting GPS tracks](#) from McCune KB et al. (2020)
- Jay [processing protocol](#)

Analysis plan

We ran all analyses in R [current version 4.5.0; R Core Team (2017)] using the following R packages: rethinking (McElreath, 2020a), rstan (Stan Development Team, 2020), cmdstanr (Gabry & Češnovar, 2022), knitr (Xie, 2018), and irr (Gamer et al., 2012). We describe below the Bayesian models we developed to address each of the specific research questions. All Bayesian models were developed using McElreath (2020b) as a guide. Code for running these models is available in supplementary materials S4.

J.Q1 Do jay populations in human modified areas differ in baseline behavioral flexibility (Reversal 1 performance) compared to populations in natural areas?

The model

We used the reversal learning Bayesian model in Logan CJ et al. (2020) to simulate and analyze population differences in reversal learning, and calculate our ability to detect differences between populations. The model “accounts for every choice made in the reversal learning experiment and updates the probability of

choosing either option after the choice was made depending on whether that choice contained a food reward or not. It does this by updating three main components for each choice: an attraction score, a learning rate (ϕ), and a rate of deviating from learned attractions (λ)” (Logan CJ et al., 2020).

Equation 1 (attraction and ϕ): $A_{i,j,t+1} = (1 - \phi_j)A_{i,j,t} + \phi_j\pi_{i,j,t}$

Equation 1 “tells us how attractions to different behavioral options $A_{i,j,t+1}$ (i.e., how preferable option i is to the bird j at time $t + 1$) change over time as a function of previous attractions $A_{i,j,t}$ and recently experienced payoffs $\pi_{i,j,t}$ (i.e., whether they received a reward in a given trial or not). Attraction scores thus reflect the accumulated learning history up to this point. The (bird-specific) parameter ϕ_j describes the weight of recent experience. The higher the value of ϕ_j , the faster the bird updates their attraction. It thus can be interpreted as the *learning or updating rate of an individual*. A value of $\phi_j = 0.04$, for example, means that receiving a single reward for one of the two options will shift preferences by 0.02 from initial 0.5-0.5 attractions, a value of $\phi_j = 0.06$ will shift preferences by 0.03 and so on” (Blaisdell et al., 2021).

Equation 2 (λ): $P(i)_{t+1} = \frac{\exp(\lambda_j A_{i,j,t})}{\sum_{m=1}^2 \exp(\lambda_j A_{m,j,t})}$.

Equation 2 “expresses the probability an individual j chooses option i in the next round, $t + 1$, based on the latent attractions. The parameter λ_j represents the *rate of deviating from learned attractions* of an individual (also called inverse temperature). It controls how sensitive choices are to differences in attraction scores. As λ_j gets larger, choices become more deterministic, as it gets smaller, choices become more exploratory (random choice if $\lambda_j = 0$). For instance, if an individual has a 0.6-0.4 preference for option A, a value of $\lambda_j = 3$ means they choose A 65% of the time, a value of $\lambda_j = 10$ means they choose A 88% of the time and a value of $\lambda_j = 0.5$ means they choose A only 53% of the time” (Blaisdell et al., 2021).

We used the c values as the response variable in the Bayesian model to examine whether there were differences in flexibility between the habitats:

$$y \sim \alpha[\text{habitat}]$$

y is the response variable (ϕ_j and λ_j , which are extracted from the correct and incorrect choices in the serial reversals). There is one intercept, α , per habitat (suburban or natural) and we will estimate the habitat’s average and standard deviation of the response variable.

J.Q2 Are disturbance-resilient jays more flexible than disturbance-resistant jays?

We were not able to collect data on disturbance-resilient species within the timeframe of this study and the funding of our grant.

J.Q3 Does manipulating behavioral flexibility alter the number of microhabitats used?

The model

Bayesian model with a normal distribution:

$$\text{habitatuse} \sim \alpha[\text{ind}] + \beta[\text{ind}] * \text{before}$$

habitatuse is the response variable: the total number of different microhabitats used per individual. There was one intercept, α , and one slope β per individual, which was estimated for the two conditions, before (and after) the manipulation. ID is nested within condition as a random effect because there was more than one data point per individual: each individual has a data point in the before condition and in the after condition. A normal distribution was used because the response variable is a sum without an expected skew to the curve (see Figure 10.6 in McElreath, 2020b).